

Introduction to Cloud Computing

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Agenda

- What is Cloud Computing?
- Use Cases
- Benefits of Cloud Computing
- Cloud Service Models
- Deployment Models
- Cloud Providers
- Core Services
- Shared Responsibility Model
- Demo

What is Cloud Computing?

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases.

Data Center

A data center is a physical location that stores computing machines and their related hardware equipment.



Use Cases

- Application Development & Deployment
- Storage & backup
- Big Data & Analytics
- Disaster recovery
- Artificial Intelligence & Machine Learning

Benefits of Cloud Computing

- Scalability
- Reliability
- Cost-efficiency
- Flexibility
- Disaster recovery

Cloud Service Models

- **Infrastructure as a Service (IaaS)**
IaaS contains the basic building blocks for cloud IT. It typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space.
- **Platform as a Service (PaaS)**
PaaS removes the need for you to manage underlying infrastructure (usually hardware and operating systems), and allows you to focus on the deployment and management of your applications.
- **Software as a Service (SaaS)**
SaaS provides you with a complete product that is run and managed by the service provider.

Deployment Models

- **Cloud (public)**
A cloud-based application is fully deployed in the cloud and all parts of the application run in the cloud.
- **Private (on-premises)**
organizations supply their own resources. That means the business is typically responsible for owning and maintaining the hardware, software, and networking infrastructure.
- **Hybrid**
A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud.
- **Community**
A cloud infrastructure shared by several organizations with common interests.

Checkpoint

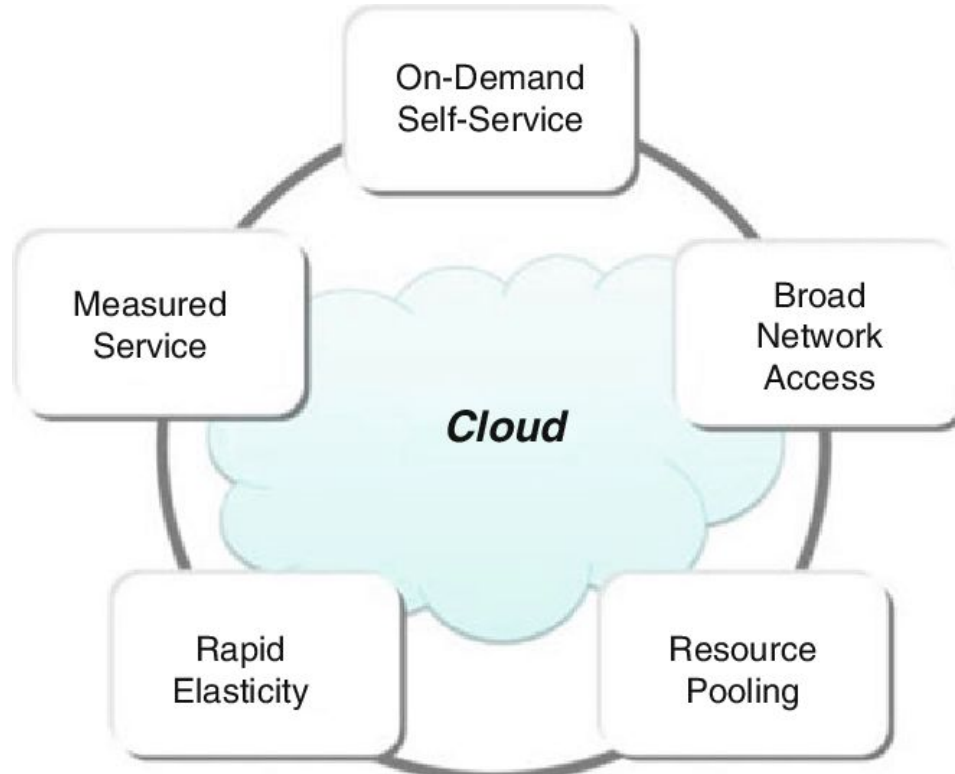
Cloud Providers

A cloud service provider, or CSP, is an IT company that provides on-demand, scalable computing resources like computing power, data storage, or applications over the internet.

Examples:

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Microsoft Azure

Characteristics of cloud provider



Characteristics of cloud provider Cont'd

- **On-demand self-services:**

The Cloud computing services does not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.

- **Broad network access:**

The Computing services are generally provided over standard networks and heterogeneous devices.

- **Rapid elasticity:**

The Computing services should have IT resources that are able to scale out and in quickly and on a need basis.

Characteristics of cloud provider Cont'd

- **Resource pooling:**

The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.

- **Measured service:**

The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used.

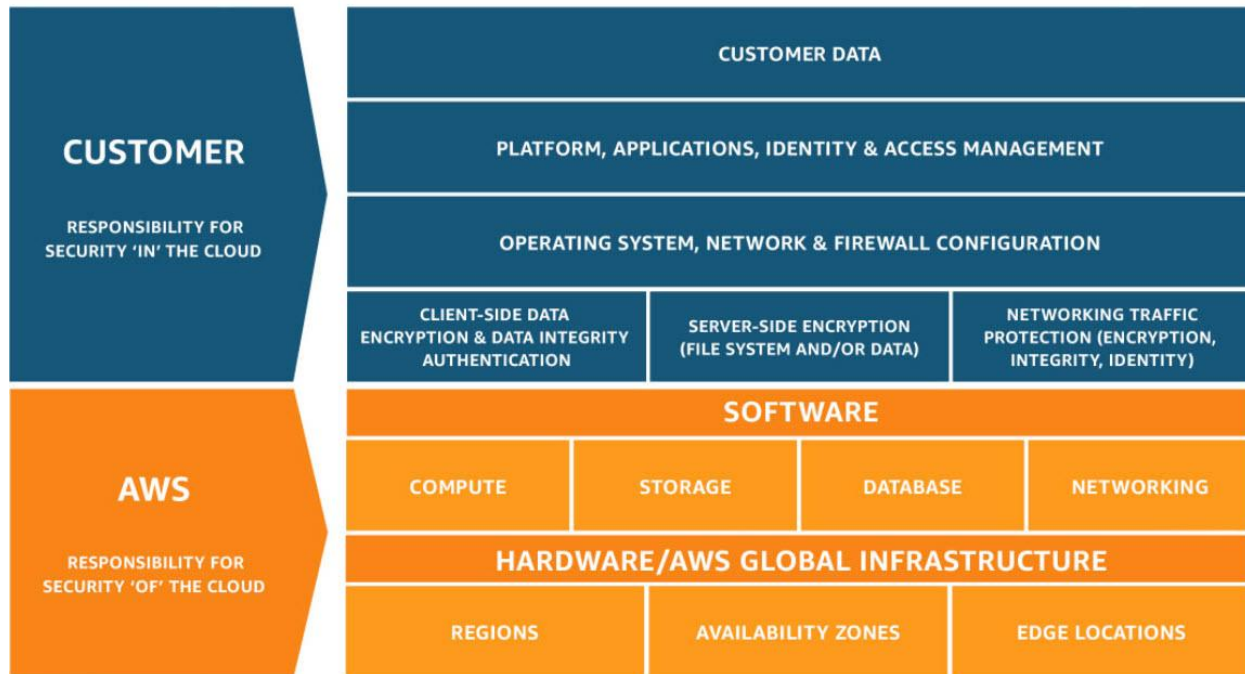
Core Services

Provider	Compute	Storage	Database	Networking
AWS	EC2	S3	RDS	VPC
GCP	GCE	GCS	Cloud SQL	VPC
Azure	VMs	Blob	SQL DB	Virtual Network

Shared Responsibility Model

The Shared Responsibility Model is a security and compliance framework that outlines the responsibilities of cloud service providers (CSPs) and customers for securing every aspect of the cloud environment, including hardware, infrastructure, endpoints, data, configurations, settings, operating system (OS), network controls and access rights.

AWS Shared Responsibility Model



Demo

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Thank You